The E_{47} spectral kernel

Three spin-2 factors give 125 coordinates. Two Casimir shells retain 47.

Shell selection

$C = (J_1 + J_2 + J_3)^2$ • eigenvalue $c_j = J(J + 1)$ • trace mean 18, deviation 12

$K = (C - 6I)(C - 30I)$ • shell values $18 - 12 = 6$ and $18 + 12 = 30$

Spin J	c_j	Copies	States	K^2 value
0	0	1	1	32,400
1	2	3	9	12,544
2	6	5	25	0
3	12	4	28	11,664
4	20	3	27	19,600
5	30	2	22	0
6	42	1	13	186,624

$$25 + 22 = 47$$

Fixed space $5V_2 \oplus 2V_5$

Orthogonal complement $125 - 47 = 78$

Exact contraction

$\Gamma = I - \varepsilon K^2$ $\varepsilon = 2/(11{,}664 + 186{,}624) = 1/99{,}144$

The endpoint multipliers are $+15/17$ and $-15/17$. Both attain the bound.

$$\|\Gamma^n - P\|_2 = (15/17)^n$$

P projects orthogonally onto $\ker K = \mathrm{Fix}(\Gamma)$. Here n counts iterations.

Arithmetic corrections

$\alpha = (125/47)^2 \approx 7.07333635$. At input spin 44, the upper shell has $J = 99$.

Upper-shell integer spins require $2j + 1 = F_k$ with k odd and $3 \nmid k$.

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